# AVH Series

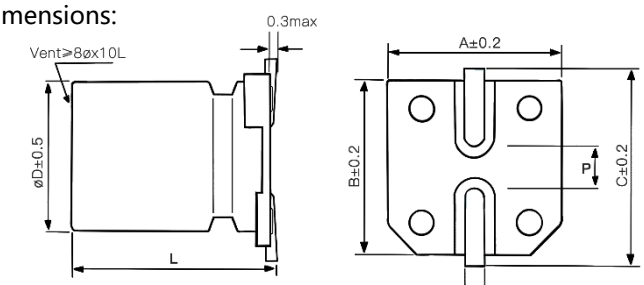
Features

- ϕ6.3 ~ ϕ12.5, 125℃, 1000~2000 hours assured
- For automobile modules and other high temperature applications
- Designed for reflow soldering
- Designed for surface mounting on high-density PCB
- Vibration resistant structure
- RoHS compliant, REACH&SVHC compliant
- AEC-Q200 compliant, Please contact Jarson for more details, test data, information

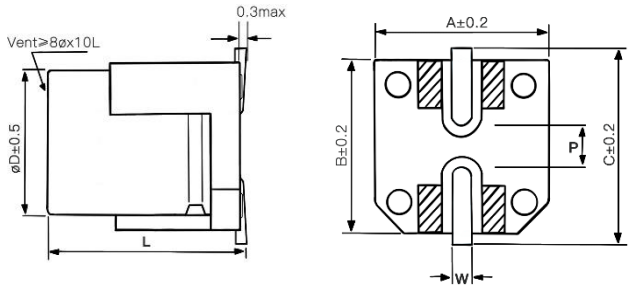
Marking color: Black

| Specifications | | | | | | | |
|---|---|---|---|---|---|---|---|
| Category temp. range | –40℃ to +125℃ | | | | | | |
| Capacitance tolerance | ±20% (120 Hz / +20 ℃) | | | | | | |
| Leakage current | I ≦ 0.01 CV or 3 μA whichever is greater (after 2 minutes) | | | | | | |
| Tan δ | Please see the attached characteristics list | | | | | | |
| Characteristics at low temperature(Impedance ratio at 120 Hz) | Rated voltage (V) | 10 | 16 | 25 | 35 | 50 | 63 |
| | Z (-25 ℃) / Z (+20 ℃) | 6 | 5 | 4 | 3 | 3 | 3 |
| | Z (-40 ℃) / Z (+20 ℃) | 12 | 8 | 6 | 4 | 4 | 4 |
| Endurance | After applying rated working voltage for 1000/2000 hours at +125 ℃ ± 2 ℃, and then being stabilized at +20 ℃, capacitors shall meet the following limits. | | | | | | |
| | Test Time | ϕD ≦ 8x6.5mm: 1000H , ϕD ≧ 8mm: 2000H | | | | | |
| | Capacitance change | Within ±30% of the initial value | | | | | |
| | Dissipation factor (tan δ) | Less than 300% of the initial value | | | | | |
| | Leakage current | Within the initial limit | | | | | |
| Shelf life | After storage for 1000 h at +125 ℃ ± 2 ℃ with no voltage applied and then being stabilized at +20 ℃, capacitors shall meet the limits specified in endurance. | | | | | | |
| Resistance to soldering heat | After reflow soldering and then being stabilized at +20 ℃, capacitors shall meet the following limits. | | | | | | |
| | Capacitance change | Within ±10% of the initial value | | | | | |
| | Dissipation factor (tan δ) | Within the initial limit | | | | | |
| | Leakage current | Within the initial limit | | | | | |
| Ripple Current and Frequency Multipliers | Frequency | 50Hz | 120Hz | 1kHz | 10kHz≦ | | |
| | Multiplier | 0.5 | 0.65 | 0.85 | 1.0 | | |

Dimensions:




Vibration resistant structure:

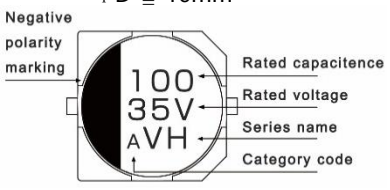



| Dimensions | | | | | | Unit: mm |
|---|---|---|---|---|---|---|
| ϕD | L | A | B | C | W | P±0.2 |
| 6.3 | 5.7±0.3 | 6.6 | 6.6 | 7.3 | 0.5~0.8 | 2.0 |
| 6.3 | 7.7±0.3 | 6.6 | 6.6 | 7.3 | 0.5~0.8 | 2.0 |
| 8 | 6.5±0.5 | 8.3 | 8.3 | 9.1 | 0.7~1.3 | 3.1 |
| 8 | 10.5±0.5 | 8.3 | 8.3 | 9.1 | 0.7~1.3 | 3.1 |
| 10 | 10.5±0.5 | 10.3 | 10.3 | 11.1 | 0.7~1.3 | 4.4 |
| 12.5 | 13.5±0.5 | 13.0 | 13.0 | 14.0 | 1.1~1.4 | 4.4 |
| 12.5 | 16±0.5 | 13.0 | 13.0 | 14.0 | 1.1~1.4 | 4.4 |

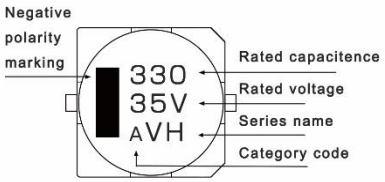
: Aid electrode

Marking:

ΦD ≦ 10mm




ΦD ≧ 12.5mm




**Characteristics list**

| Rated voltage (V) | Capacitance (±20%) (μF) | Case size | | Specification | | | Part Number④ | Taping&Reel |
|---|---|---|---|---|---|---|---|---|
| | | øD (mm) | L (mm) | Rated ripple current① (mA rms) | Imp.② (Ω) | tan δ③ | | MPQ (pcs/reel) |
| 10 | 47 | 6.3 | 5.7 | 70 | 1.60 | 0.30 | AVH1A470M0606 | 1000 |
| | 68 | 6.3 | 5.7 | 70 | 1.60 | 0.30 | AVH1A680M0606 | 1000 |
| | 100 | 6.3 | 7.7 | 100 | 1.00 | 0.30 | AVH1A101M0607 | 1000 |
| | | 8 | 6.5 | 100 | 1.00 | 0.30 | AVH1A101M0806 | 1000 |
| | 220 | 8 | 10.5 | 200 | 0.50 | 0.30 | AVH1A221M0810 | 500 |
| | 330 | 8 | 10.5 | 200 | 0.50 | 0.30 | AVH1A331M0810 | 500 |
| | 470 | 10 | 10.5 | 280 | 0.30 | 0.30 | AVH1A471M1010 | 500 |
| | 680 | 12.5 | 13.5 | 750 | 0.12 | 0.30 | AVH1A681M1313 | 200 |
| | 1000 | 12.5 | 13.5 | 750 | 0.12 | 0.30 | AVH1A102M1313 | 200 |
| | 1500 | 12.5 | 13.5 | 750 | 0.12 | 0.32 | AVH1A152M1313 | 200 |
| 16 | 33 | 6.3 | 5.7 | 70 | 1.60 | 0.23 | AVH1C330M0606 | 1000 |
| | 47 | 6.3 | 7.7 | 100 | 1.00 | 0.23 | AVH1C470M0607 | 1000 |
| | 100 | 6.3 | 7.7 | 100 | 1.00 | 0.23 | AVH1C101M0607 | 1000 |
| | | 8 | 6.5 | 100 | 1.00 | 0.23 | AVH1C101M0806 | 1000 |
| | 220 | 6.3 | 7.7 | 100 | 1.00 | 0.23 | AVH1C221M0607 | 1000 |
| | | 8 | 10.5 | 200 | 0.50 | 0.23 | AVH1C221M0810 | 500 |
| | | 10 | 10.5 | 280 | 0.30 | 0.23 | AVH1C221M1010 | 500 |
| | 330 | 10 | 10.5 | 280 | 0.30 | 0.23 | AVH1C331M1010 | 500 |
| | | 12.5 | 13.5 | 750 | 0.12 | 0.23 | AVH1C331M1313 | 200 |
| | 470 | 10 | 10.5 | 280 | 0.30 | 0.23 | AVH1C471M1010 | 500 |
| | | 12.5 | 13.5 | 750 | 0.12 | 0.23 | AVH1C471M1313 | 200 |
| | 680 | 12.5 | 13.5 | 750 | 0.12 | 0.23 | AVH1C681M1313 | 200 |
| | 1000 | 12.5 | 13.5 | 750 | 0.12 | 0.23 | AVH1C102M1313 | 200 |
| 25 | 33 | 6.3 | 5.7 | 70 | 1.60 | 0.18 | AVH1E330M0606 | 1000 |
| | 47 | 6.3 | 7.7 | 100 | 1.00 | 0.18 | AVH1E470M0607 | 1000 |
| | | 8 | 6.5 | 100 | 1.00 | 0.18 | AVH1E470M0806 | 1000 |
| | 100 | 6.3 | 7.7 | 100 | 1.00 | 0.18 | AVH1E101M0607 | 1000 |
| | | 8 | 6.5 | 100 | 1.00 | 0.18 | AVH1E101M0806 | 1000 |
| | | 8 | 10.5 | 200 | 0.50 | 0.18 | AVH1E101M0810 | 500 |
| | 220 | 8 | 10.5 | 200 | 0.50 | 0.18 | AVH1E221M0810 | 500 |
| | | 10 | 10.5 | 280 | 0.30 | 0.18 | AVH1E221M1010 | 500 |
| | 330 | 10 | 10.5 | 280 | 0.30 | 0.18 | AVH1E331M1010 | 500 |
| | | 12.5 | 13.5 | 750 | 0.12 | 0.18 | AVH1E331M1313 | 200 |
| | 470 | 12.5 | 13.5 | 750 | 0.12 | 0.18 | AVH1E471M1313 | 200 |
| | 680 | 12.5 | 13.5 | 750 | 0.12 | 0.18 | AVH1E681M1313 | 200 |

① Rated ripple current (100kHz / +125℃) ② Impedance (100kHz / +20℃) ③ tan δ (120Hz / +20℃)

※Please refer to the page of reflow conditions for reflow profile.

## Characteristics list

| Rated voltage (V) | Capacitance (±20%) (μF) | Case size | | Specification | | | Part Number④ | Taping&Reel |
|---|---|---|---|---|---|---|---|---|
| | | øD (mm) | L (mm) | Rated ripple current① (mA rms) | Imp.② (Ω) | tan δ③ | | MPQ (pcs/reel) |
| 35 | 22 | 6.3 | 5.7 | 70 | 1.60 | 0.16 | AVH1V220M0606 | 1000 |
| | 33 | 6.3 | 7.7 | 100 | 1.00 | 0.16 | AVH1V330M0607 | 1000 |
| | | 8 | 6.5 | 100 | 1.00 | 0.16 | AVH1V330M0806 | 1000 |
| | 47 | 6.3 | 7.7 | 100 | 1.00 | 0.16 | AVH1V470M0607 | 1000 |
| | | 8 | 6.5 | 100 | 1.00 | 0.16 | AVH1V470M0806 | 1000 |
| | | 8 | 10.5 | 200 | 0.50 | 0.16 | AVH1V470M0810 | 500 |
| | 100 | 6.3 | 7.7 | 100 | 1.00 | 0.16 | AVH1V101M0607 | 1000 |
| | | 8 | 10.5 | 200 | 0.50 | 0.16 | AVH1V101M0810 | 500 |
| | | 10 | 10.5 | 280 | 0.30 | 0.16 | AVH1V101M1010 | 500 |
| | 220 | 10 | 10.5 | 280 | 0.30 | 0.16 | AVH1V221M1010 | 500 |
| | | 12.5 | 13.5 | 750 | 0.12 | 0.16 | AVH1V221M1313 | 200 |
| | 330 | 10 | 10.5 | 280 | 0.30 | 0.16 | AVH1V331M1010 | 500 |
| | | 12.5 | 13.5 | 750 | 0.12 | 0.16 | AVH1V331M1313 | 200 |
| | 470 | 12.5 | 13.5 | 750 | 0.12 | 0.16 | AVH1V471M1313 | 200 |
| 50 | 10 | 6.3 | 7.7 | 85 | 1.60 | 0.15 | AVH1H100M0607 | 1000 |
| | | 8 | 6.5 | 85 | 1.60 | 0.15 | AVH1H100M0806 | 1000 |
| | 22 | 6.3 | 7.7 | 85 | 1.60 | 0.15 | AVH1H220M0607 | 1000 |
| | | 8 | 6.5 | 85 | 1.60 | 0.15 | AVH1H220M0806 | 1000 |
| | 33 | 6.3 | 7.7 | 85 | 1.60 | 0.15 | AVH1H330M0607 | 1000 |
| | | 8 | 6.5 | 85 | 1.60 | 0.15 | AVH1H330M0806 | 1000 |
| | 47 | 8 | 10.5 | 160 | 0.75 | 0.15 | AVH1H470M0810 | 500 |
| | | 10 | 10.5 | 240 | 0.50 | 0.15 | AVH1H470M1010 | 500 |
| | 100 | 10 | 10.5 | 240 | 0.50 | 0.15 | AVH1H101M1010 | 500 |
| | | 12.5 | 13.5 | 550 | 0.23 | 0.15 | AVH1H101M1313 | 200 |
| | 220 | 12.5 | 13.5 | 550 | 0.23 | 0.15 | AVH1H221M1313 | 200 |
| | 330 | 12.5 | 13.5 | 550 | 0.23 | 0.15 | AVH1H331M1313 | 200 |
| 63 | 10 | 6.3 | 7.7 | 60 | 2.20 | 0.13 | AVH1J100M0607 | 1000 |
| | | 8 | 6.5 | 60 | 2.20 | 0.13 | AVH1J100M0806 | 1000 |
| | 22 | 8 | 10.5 | 100 | 1.00 | 0.13 | AVH1J220M0810 | 500 |
| | 33 | 8 | 10.5 | 100 | 1.00 | 0.13 | AVH1J330M0810 | 500 |
| | | 10 | 10.5 | 150 | 0.80 | 0.13 | AVH1J330M1010 | 500 |
| | 47 | 8 | 10.5 | 100 | 1.00 | 0.13 | AVH1J470M0810 | 500 |
| | | 10 | 10.5 | 150 | 0.80 | 0.13 | AVH1J470M1010 | 500 |
| | 100 | 10 | 10.5 | 150 | 0.80 | 0.13 | AVH1J101M1010 | 500 |
| | | 12.5 | 13.5 | 450 | 0.26 | 0.13 | AVH1J101M1313 | 200 |
| | 220 | 12.5 | 13.5 | 450 | 0.26 | 0.13 | AVH1J221M1313 | 200 |

① Rated ripple current (100kHz / +125℃) ② Impedance (100kHz / +20℃) ③ tan δ (120Hz / +20℃)

※Please refer to the page of reflow conditions for reflow profile.

Part Number System:

| SMD E-Cap | AVH series | 16V | 220μF | ±20 % | 6.3 φ x7.7L | |
|---|---|---|---|---|---|---|
| **A** | **VH** | **1C** | **221** | **M** | **0607** | |
| Category | Series name | Rated voltage | Capacitance | Tolerance | Case Size | Supplement |

④ Note: For more details, please refer to "Part Number System - SMD Type"